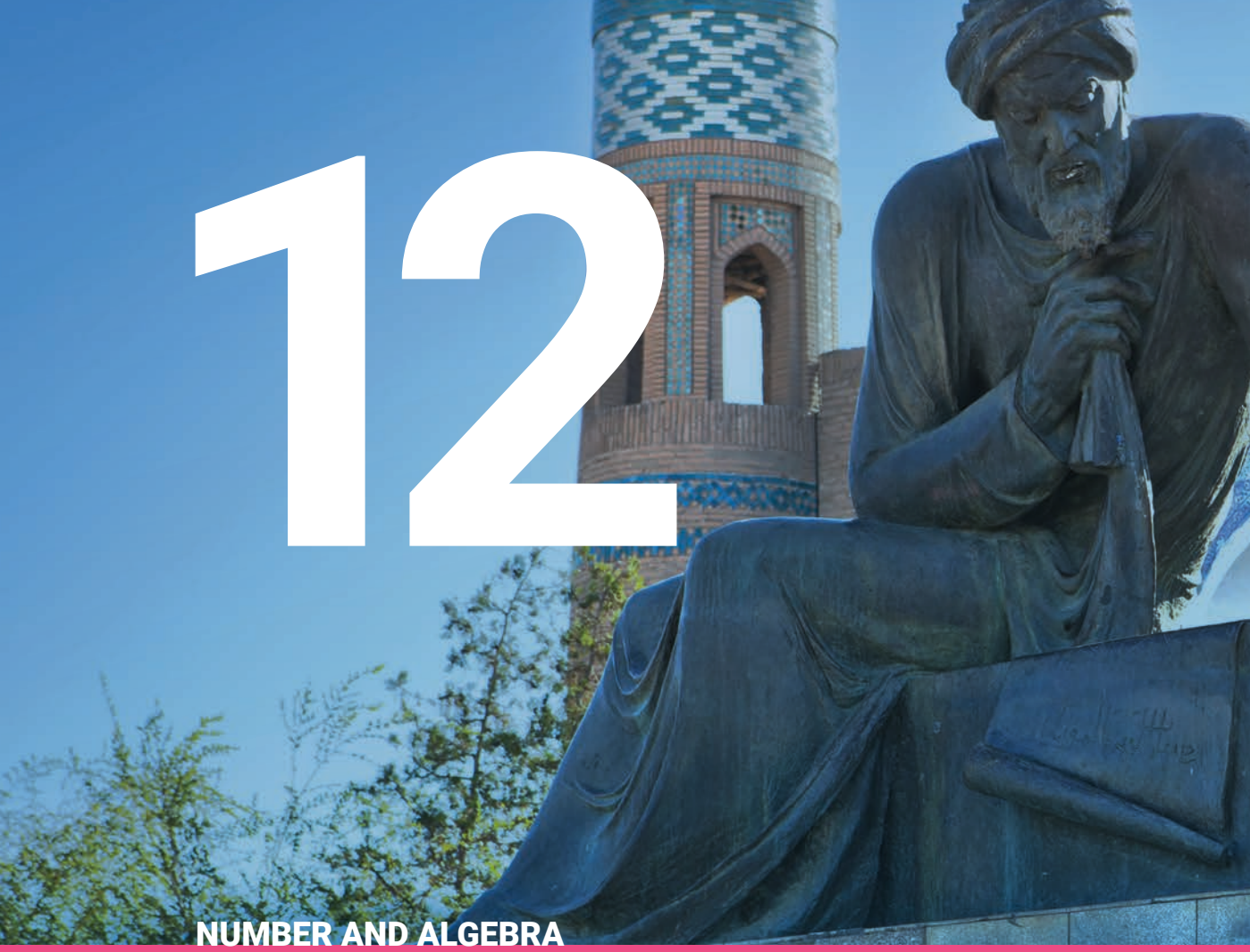


# 12

NUMBER AND ALGEBRA

# PRODUCTS AND FACTORS

## OPTIONAL PATH TOPIC
## RECOMMENDED FOR STAGE 6 MATHEMATICS ADVANCED

In 825 CE, the Persian mathematician al-Khwarizmi used the Arabic word ‘al-jabr’ to describe the process of adding equal quantities to both sides of an equation. When al-Khwarizmi’s book was translated into Latin and introduced to Europe, ‘al-jabr’ became ‘algebra’ and the word was adopted as the name for the branch of mathematics that uses formulas to describe patterns and relationships in our world. The photo above shows a monument of al-Khwarizmi located in Khiva, Uzbekistan.




## Chapter outline

| | Working mathematically | | | | |
|---|---|---|---|---|---|
| **12.01** Perfect squares* | U | F | | | |
| **12.02** Difference of 2 squares* | U | F | | R | C |
| **12.03** Mixed expansions* | U | F | | R | C |
| **12.04** Factorising special binomial products* | U | F | | | |
| **12.05** Factorising quadratic expressions $ax^2 + bx + c$* | U | F | | R | |
| **12.06** Mixed factorisations* | U | F | | R | |
| **12.07** Factorising algebraic fractions* | U | F | | R | |

*PATH

## Wordbank

**binomial** An algebraic expression that consists of 2 terms, for example, $4a + 9$, $3 - y$, $x^2 - 4x$

**binomial product** An algebraic expression showing 2 or more binomials multiplied together, for example, $(x + 9)(3x - 4)$

**factorise** To rewrite an expression with grouping symbols, by taking out the highest common factor; factorising is the opposite of expanding, for example, $9r^2 + 36r$ factorised is $9r(r + 4)$

**highest common factor (HCF)** The largest term that is a factor of 2 or more terms, for example, the HCF of $9r^2 + 36r$ is $9r$

**perfect square** A square number or an algebraic expression that represents one, for example, 64, $(x + 9)^2$

**quadratic expression** An algebraic expression in which the highest power of the variable is 2, for example, $2x^2 + 5x - 3$ or $x^2 + 2$

**quadratic trinomial** An algebraic expression that consists of 3 terms, for example, $x^2 + 2x + 6$

**Quiz**
Wordbank 12

**U** = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

**Videos (6):**

**12.01** Special binomial products

**12.02** Difference of 2 squares • Special binomial products

**12.04** Difference of 2 squares

**12.05** Factorising quadratic expressions (Advanced) • Factorising quadratic expressions 2 • Advanced algebra

**Quizzes (2):**

- Wordbank 12
- SkillCheck 12

**Worksheets (4):**

**12.02** Special products

**12.03** Mixed expansions • Algebra 6

**12.07** Simplifying algebraic fractions

**Puzzles (10):**

**12.01** Factorominoes

**12.02** Enough time

**12.03** Products and factors squaresaw • Expanding brackets

**12.04** Grouping • Difference of 2 perfect squares

**12.05** Products and factors squaresaw • Perfect squares • Factorominoes • Factorising puzzle (Advanced) • Trinominoes

**12.06** Mixed factorisations • Factorominoes • Products and factors squaresaw

**Technology (1):**

**12.05** Factorising trinomials

**Spreadsheet (1):**

**12.05** Factorising trinomials

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

## In this chapter you will:

- (PATH) recognise and expand special binomial products that are perfect squares or the difference of 2 squares
- (PATH) factorise an expression with 4 terms by grouping in pairs
- (PATH) factorise non-monic quadratic expressions, including those that are perfect squares or the difference of 2 squares
- (PATH) factorise and simplify algebraic fractions

PATH

Quiz
SkillCheck 12

## SkillCheck ANSWERS ON P. 635

**1** Expand each binomial product.

**a** $(x + 1)(x + 7)$

**b** $(d + 3)(d - 4)$

**c** $(t - 5)(t + 5)$

**d** $(y + 2)(5y - 6)$

**e** $(7y - 3)(7y - 3)$

**f** $(5p + 4)(2p - 7)$

**2** Factorise each quadratic expression.

**a** $y^2 + 10y + 25$

**b** $x^2 - 21x + 20$

**c** $n^2 + 8n - 33$

**d** $a^2 - 11a + 28$

**e** $m^2 - 5m - 84$

**f** $p^2 + 3p - 54$

# Perfect squares

12.01

We know how to expand **binomial products** such as $(k + 3)(k - 7)$. Now we will examine the special binomial product where a binomial is multiplied by itself, for example,

$(x + 4)(x + 4) = (x + 4)^2$ or $(2a - 9)(2a - 9) = (2a - 9)^2$.

16, 49, $v^2$ and $(y + 5)^2$ are called **perfect squares** because they are square numbers.

PATH



Puzzle
Factorominoes



## Perfect squares

The formulas for expanding the perfect square of a binomial are:

$$(a + b)^2 = a^2 + 2ab + b^2$$
$$(a - b)^2 = a^2 - 2ab + b^2$$

**Proof**

$(a + b)^2 = (a + b)(a + b)$
$= a(a + b) + b(a + b)$
$= a^2 + ab + ba + b^2$
$= a^2 + 2ab + b^2$

$(a - b)^2 = (a - b)(a - b)$
$= a(a - b) - b(a - b)$
$= a^2 - ab - ba + b^2$
$= a^2 - 2ab + b^2$

## Example 1

Copy and complete the expansion of each perfect square.

**a** $(x + 7)^2 = x^2 + ____ + 49$

**b** $(y - 6)^2 = y^2 - 12y + ____$

**c** $(5g + 9)^2 = ____ + 90g + 81$

**d** $(3d - 5)^2 = 9d^2 - ____ + 25$

### Solution

**a** In the expansion,

$_____ = 2 \times x \times 7$ Doubling the product of the 2 terms
$= 14x$
$\therefore (x + 7)^2 = x^2 + 14x + 49$

**b** In the expansion,

$_____ = 6^2$ The second term squared
$= 36$
$\therefore (y - 6)^2 = y^2 - 12y + 36$

**c** $_____ = (5g)^2$ The first term squared
$= 25g^2$
$\therefore (5g + 9)^2 = 25g^2 + 90g + 81$

**d** $_____ = 2 \times 3d \times 5$ Doubling the product of the 2 terms
$= 30d$
$\therefore (3d - 5)^2 = 9d^2 - 30d + 25$

PATH

Video
Special binomial products

## Example 2

Expand each perfect square.

**a** $(n-5)^2$ **b** $(k+4)^2$ **c** $(3y-8)^2$

### Solution

**a** $(n-5)^2 = n^2 - 2 \times n \times 5 + 5^2$
$= n^2 - 10n + 25$

1st term squared – double product + 2nd term squared

**b** $(k+4)^2 = k^2 + 2 \times k \times 4 + 4^2$
$= k^2 + 8k + 16$

**c** $(3y-8)^2 = (3y)^2 - 2 \times 3y \times 8 + 8^2$
$= 9y^2 - 48y + 64$

PATH

**EXERCISE 12.01** ANSWERS ON P. 635

## Perfect squares U F

EXAMPLE 1

**1** Copy and complete the expansion of each perfect square.

**a** $(x+10)^2 = x^2 + _____ + 100$ **b** $(m-8)^2 = _____ - 16m + 64$

**c** $(p-t)^2 = p^2 - 2pt + _____$ **d** $(h+4)^2 = h^2 _____ + 16$

**e** $(k-9)^2 = k^2 _____ + 81$ **f** $(8+5f)^2 = 64 _____ + 25f^2$

**g** $(2d+3)^2 = _____ + _____ + 9$ **h** $(6a+1)^2 = _____ + 12a + _____$

EXAMPLE 2

**2** Expand each perfect square.

**a** $(m+9)^2$ **b** $(u+3)^2$ **c** $(y-6)^2$

**d** $(8+k)^2$ **e** $(5-h)^2$ **f** $(7+k)^2$

**g** $(f+20)^2$ **h** $(q-11)^2$ **i** $(10+t)^2$

**j** $(x-w)^2$ **k** $(a+g)^2$ **l** $(2m-3)^2$

**m** $(5x-6)^2$ **n** $(9a+2)^2$ **o** $(3e-4)^2$

**p** $(5+7b)^2$ **q** $(4-5p)^2$ **r** $(11-2c)^2$

**s** $(10g+3)^2$ **t** $(3k+11)^2$ **u** $(5+2v)^2$

**3** Expand each perfect square.

**a** $(7h+2k)^2$ **b** $(8a-3y)^2$ **c** $(xy+z)^2$

**d** $\left(1+\frac{1}{y}\right)^2$ **e** $\left(k-\frac{1}{k}\right)^2$ **f** $\left(w+\frac{3}{w}\right)^2$

**4** Use expansion to evaluate each square number without using a calculator.

**a** $21^2 = (20+1)^2$ **b** $45^2 = (40+5)^2$ **c** $29^2 = (30-1)^2$

**d** $59^2 = (60-1)^2$ **e** $102^2 = (100+2)^2$ **f** $98^2 = (100-2)^2$

Foundation Standard Complex


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# Difference of 2 squares

12.02

The binomial product $(a + b)(a - b)$ has a special pattern when expanded.

## Difference of 2 squares

$$(a + b)(a - b) = a^2 - b^2$$

The answer is called the **difference of 2 squares**.

**Proof:**

$$\begin{aligned}(a - b)(a + b) &= a(a + b) - b(a + b) \\ &= a^2 + ab - ba - b^2 \\ &= a^2 - b^2 \qquad + ab - ba = 0\end{aligned}$$

When the sum of 2 terms is multiplied by their difference, the answer is the square of the first term minus the square of the second term (the difference of 2 squares).

## Example 3

Expand each expression.

**a** $(d + 3)(d - 3)$

**b** $(2 + r)(2 - r)$

**c** $(7x + 2)(7x - 2)$

**d** $(4k - 5p)(4k + 5p)$

### Solution

**a** $(d + 3)(d - 3) = d^2 - 3^2$
$= d^2 - 9$

**b** $(2 + r)(2 - r) = 2^2 - r^2$
$= 4 - r^2$

**c** $(7x + 2)(7x - 2) = (7x)^2 - 2^2$
$= 49x^2 - 4$

**d** $(4k - 5p)(4k + 5p) = (4k)^2 - (5p)^2$
$= 16k^2 - 25p^2$

PATH

12.02

**Worksheet**
Special products

**Puzzle**
Enough time

**Videos**
Difference of 2 squares

Special binomial products

PATH

## EXERCISE 12.02 ANSWERS ON P. 635

### Difference of 2 squares U F R C

EXAMPLE 3

**1** Expand each expression.

| | | | | | |
|---|---|---|---|---|---|
| **a** | $(m + 5)(m - 5)$ | **b** | $(c - 10)(c + 10)$ | **c** | $(a + 12)(a - 12)$ |
| **d** | $(6 - y)(6 + y)$ | **e** | $(8 - m)(8 + m)$ | **f** | $(p + 1)(p - 1)$ |
| **g** | $(5 + e)(5 - e)$ | **h** | $(v + 11)(v - 11)$ | **i** | $(w - 3)(w + 3)$ |
| **j** | $(x - 10)(x + 10)$ | **k** | $(q + 7)(q - 7)$ | **l** | $(9 - g)(9 + g)$ |
| **m** | $(b - 2)(b + 2)$ | **n** | $(15 - r)(15 + r)$ | **o** | $(d + 13)(d - 13)$ |

**2** Expand each expression.

| | | | | | |
|---|---|---|---|---|---|
| **a** | $(2h - 3)(2h + 3)$ | **b** | $(5r + 4)(5r - 4)$ | **c** | $(5b + 8)(5b - 8)$ |
| **d** | $(4p - 7)(4p + 7)$ | **e** | $(3 - 8k)(3 + 8k)$ | **f** | $(7x - 5)(7x + 5)$ |
| **g** | $(2 + 9m)(2 - 9m)$ | **h** | $(9k - 4l)(9k + 4l)$ | **i** | $(7n + 8m)(7n - 8m)$ |
| **j** | $(4g - 5h)(4g + 5h)$ | **k** | $(7u + 3w)(7u - 3w)$ | **l** | $(11a + 3b)(11a - 3b)$ |
| **m** | $\left(a+\frac{1}{a}\right)\left(a-\frac{1}{a}\right)$ | **n** | $\left(\frac{w}{3}-2\right)\left(\frac{w}{3}+2\right)$ | **o** | $\left(1-\frac{1}{r}\right)\left(1+\frac{1}{r}\right)$ |

**3** Natalie's age is $p$ years. R C

- **a** What was Natalie's age last year?
- **b** What will Natalie's age be next year?
- **c** Write an expression for (Natalie's age last year) × (Natalie's age next year).
- **d** If (Natalie's age last year) × (Natalie's age next year) is equal to 48, what is Natalie's age?

**4** By expressing $31 \times 29$ as $(30 + 1)(30 - 1)$, use the difference of 2 squares to find the value of $31 \times 29$. R C

**5** Use the method of question **4** to evaluate each expression. R C

**a** $21 \times 19$ **b** $51 \times 49$ **c** $89 \times 91$ **d** $78 \times 82$

**6** Use 'sum by difference' to evaluate each expression.

| | | | | | |
|---|---|---|---|---|---|
| **a** | $15^2 - 14^2$ | **b** | $24^2 - 23^2$ | **c** | $65^2 - 35^2$ |
| **d** | $101^2 - 99^2$ | **e** | $23^2 - 17^2$ | **f** | $50^2 - 48^2$ |

Foundation Standard Complex


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# Mixed expansions 12.03

## Example 4

Expand and simplify each expression.

**a** $(4r + 5)(1 - 2r)$ **b** $(7 + 9x)^2$ **c** $(3d - 10)(3d + 10)$

**d** $(a + 6)(a - 6) + (a + 12)(a + 3)$ **e** $(m - 2)^2 - (m - 2)(m + 2)$

### Solution

**a** $(4r + 5)(1 - 2r) = 4r(1 - 2r) + 5(1 - 2r)$ Expanding

$= 4r - 8r^2 + 5 - 10r$

$= -8r^2 - 6r + 5$ Rearranging and putting the higher order terms first.

**b** $(7 + 9x)^2 = 7^2 + 2 \times 7 \times 9x + (9x)^2$ Expand the perfect square.

$= 49 + 126x + 81x^2$

$= 81x^2 + 126x + 49$ Rearranging and putting the higher order terms first.

**c** $(3d - 10)(3d + 10) = (3d)^2 - 10^2$ Expand into the difference of 2 squares

$= 9d^2 - 100$

**d** $(a + 6)(a - 6) + (a + 12)(a + 3) = a^2 - 36 + a^2 + 3a + 12a + 36$

$= 2a^2 + 15a$

**e** $(m - 2)^2 - (m - 2)(m + 2) = m^2 - 4m + 4 - (m^2 - 4)$

$= m^2 - 4m + 4 - m^2 + 4$

$= -4m + 8$

PATH

Worksheets
Mixed expansions
Algebra 6

Puzzles
Products and factors squaresaw
Expanding brackets

12.03

## EXERCISE 12.03 ANSWERS ON P. 636

### Mixed expansions U F R C

EXAMPLE 4

**1** Expand and simplify each expression.

**a** $(2m - 1)(2m + 1)$ **b** $(y + 4)(5y - 3)$ **c** $(2k - 7)^2$

**d** $(d + 9)^2$ **e** $(2e - 1)(e + 1)$ **f** $(5a + 4)(5a - 4)$

**g** $(2 - p)(p - 2)$ **h** $(10 - 6y)(10 + 6y)$ **i** $(h - 3m)^2$

**j** $(2x - 3)(y + 3)$ **k** $(11a - 4b)(11a + 4b)$ **l** $\left(u - \frac{1}{u}\right)^2$

Foundation Standard Complex

PATH

**2** Expand and simplify each expression. R C

| | | | |
|---|---|---|---|
| **a** | $(m-5)(m+5)+25$ | **b** | $6y+(y-3)^2+9$ |
| **c** | $(3x+1)(2-x)+2x+4$ | **d** | $(d+4)^2-8d+5$ |
| **e** | $16+(4k-8)(4k+8)$ | **f** | $(x-y)^2-(x+y)^2$ |
| **g** | $20a-(a-2)(a-5)+a^2$ | **h** | $2(f-2)(f+2)$ |
| **i** | $(2h+3)^2-(2h-3)(2h+3)$ | **j** | $7xy-(2x-3)(y+3)$ |

**3** Expand and simplify each expression. R C

| | | | |
|---|---|---|---|
| **a** | $(8a-1)(8a+1)-4a^2+1$ | **b** | $(n+1)^2+2n+3$ |
| **c** | $3(4-u)(4+u)+(u-12)(u+4)$ | **d** | $(2m-n)^2+(2m+n)^2$ |
| **e** | $(x-2)(x+3)-(x-2)(x+2)$ | **f** | $2(b-1)^2-(2b-1)^2$ |
| **g** | $(y+1)^2+(y+2)^2+(y+3)^2$ | **h** | $(x-3)(x+3)+(x+3)^2+(x-3)^2$ |
| **i** | $(5n+3)(5n-3)+(3n-5)(3n+5)$ | **j** | $2(a-b)(a+b)-(a+b)^2-(a-b)^2$ |

# 12.04 Factorising special binomial products

PATH

## Factorising by grouping in pairs

Puzzle
Grouping

An **algebraic expression with 4 terms** can often be factorised in pairs, that is, 2 terms at a time, to make a binomial product.

### Example 5

Factorise each expression.

**a** $3ac+2bd+2bc+3ad$

**b** $4km+6mn-6kp-9np$

**c** $10xw-6yw-10xp+6yp$

### Solution

**a** $3ac+2bd+2bc+3ad=3ac+3ad+2bd+2bc$ — Grouping into pairs for factorising

$=3a(c+d)+2b(d+c)$ — Factorising each pair

$=(c+d)(3a+2b)$ — Factorising again

**b** $4km+6mn-6kp-9np=2m(2k+3n)-3p(2k+3n)$ — Factorising each pair

$=(2k+3n)(2m-3p)$ — Factorising again

**c** $10xw-6yw-10xp+6yp=2(5xw-3yw-5xp+3yp)$ — Factorising all terms first

$=2[w(5x-3y)-p(5x-3y)]$ — Factorising each pair

$=2(5x-3y)(w-p)$ — Factorising again

Foundation Standard Complex


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# Factorising the difference of 2 squares

We have learned the formula $(a + b)(a - b) = a^2 - b^2$.

If we use this rule in reverse, then the factors of $a^2 - b^2$ are $(a + b)$ and $(a - b)$.

## The difference of 2 squares

$$a^2 - b^2 = (a + b)(a - b)$$

## Example 6

Factorise each expression.

**a** $x^2 - 4$ **b** $9 - 16b^2$ **c** $20d^2 - 5a^2$ **d** $y^3 - y$

### Solution

**a** $x^2 - 4 = x^2 - 2^2$
$= (x + 2)(x - 2)$

**b** $9 - 16b^2 = 3^2 - (4b)^2$
$= (3 + 4b)(3 - 4b)$

**c** $20d^2 - 5a^2 = 5(4d^2 - a^2)$
$= 5[(2d)^2 - a^2]$
$= 5(2d + a)(2d - a)$

**d** $y^3 - y = y(y^2 - 1)$
$= y(y + 1)(y - 1)$

**EXERCISE 12.04** ANSWERS ON P. 636

## Factorising special binomial products U F

**1** Factorise each expression.

**a** $4ab + 5bc + 4ad + 5cd$
**b** $2xy - 5wy + 2xt - 5wt$
**c** $9ac + 6bc + 12ad + 8bd$
**d** $10x^2 + 30 + x^3 + 3x$
**e** $3a^2 + 3ab + 3ac + 3bc$
**f** $6rt - 18wt + 6rp - 18wp$
**g** $14e - 21 + 2de - 3d$
**h** $hk - h^2 - 2k + 2h$
**i** $3mn - 6m + pn - 2p$
**j** $9p^2 - 27 + qp^2 - 3q$
**k** $fg - fh - 10g + 10h$
**l** $9kl - 12ml + 9kn - 12mn$
**m** $2p - 2c - p^2 + pc$
**n** $l^3 + lm^2 - 3l^2 - 3m^2$
**o** $a(x + 1) + y(x + 1) - ka - ky$
**p** $p(a - b) - 2q(a - b) + 3qp - 6q^2$

**2** Factorise each expression.

**a** $w^2 - 9$
**b** $y^2 - 36$
**c** $k^2 - 1$
**d** $m^2 - 121$
**e** $p^2 - 64$
**f** $c^2 - 100$
**g** $4e^2 - f^2$
**h** $a^2 - 9b^2$
**i** $16y^2 - 1$
**j** $4 - b^2$
**k** $25 - e^2$
**l** $1 - 16x^2$
**m** $k^2 - u^2$
**n** $49 - 16m^2$
**o** $b^2 - 121d^2$
**p** $36c^2 - 25k^2$
**q** $16 - 81h^2$
**r** $25a^2 - 64m^2$
**s** $100 - 49n^2$
**t** $121p^2 - 144q^2$
**u** $\frac{1}{4} - 25c^2$
**v** $4w^2 - \frac{1}{9}$
**w** $64h^2 - 2\frac{1}{4}$
**x** $1 - m^2 n^2$

PATH

Puzzle
Difference of 2 perfect squares

12.04

Video
Difference of 2 squares

EXAMPLE 5

EXAMPLE 6

Foundation Standard Complex

PATH

**3** Factorise each expression.

| | | | |
|---|---|---|---|
| **a** $2a^2 - 2b^2$ | **b** $7k^2 - 28$ | **c** $3 - 75u^2$ | **d** $x^3 - 49x$ |
| **e** $k - 16k^3$ | **f** $50q^2 - 2$ | **g** $3d^2 - 12v^2$ | **h** $5t^5 - 125t^3$ |
| **i** $2a^2b^2 - 2$ | **j** $x^2y^2 - x^2w^2$ | **k** $192f^2 - 108g^2$ | **l** $45d^2 - \frac{5}{4}$ |
| **m** $2x^2 - 8a^2$ | **n** $100 - 25w^2$ | **o** $1\frac{1}{4} - 80e^2$ | **p** $9c^2 - 6\frac{1}{4}$ |

**4** Factorise each expression.

| | | |
|---|---|---|
| **a** $\frac{p^2}{16} - \frac{x^2}{25}$ | **b** $x^2 - \frac{1}{9}$ | **c** $\frac{v^2}{64} - \frac{u^2}{81}$ |
| **d** $\frac{2y^2}{9} - \frac{2m^2}{121}$ | **e** $\frac{16a^2}{49} - \frac{25b^2}{4}$ | **f** $g^4 - 81$ |
| **g** $10\,000 - n^4$ | **h** $(x + y)^2 - x^2$ | **i** $(p - 2q)^2 - (2p + q)^2$ |
| **j** $\frac{x^2}{4} - \frac{y^2}{36}$ | **k** $(a + b)^2 - (a - b)^2$ | **l** $25 - (4 - x)^2$ |

# 12.05 Factorising quadratic expressions $ax^2 + bx + c$

PATH

In Chapter 4, Algebra, we factorised **quadratic trinomials** of the type $x^2 + bx + c$. For example, the factorisation of $x^2 + 6x + 8$ is $(x + 2)(x + 4)$.

We will now factorise quadratic expressions of the type $ax^2 + bx + c$, such as $6x^2 + 19x + 15$, where $x^2$ has a **coefficient**.

**Puzzles**
Products and factors squaresaw
Perfect squares
Factorominoes
Factorising puzzle (Advanced)
Trinominoes

**Technology**
Factorising trinomials

**Spreadsheet**
Factorising trinomials

## Example 7

Factorise each quadratic expression.

**a** $3g^2 + 12g - 36$ **b** $48 - 8p - p^2$

### Solution

**a** $3g^2 + 12g - 36 = 3(g^2 + 4g - 12)$ Taking out the HCF of 3 first.

$= 3(g - 2)(g + 6)$ Product = –12, sum = 4

**b** $48 - 8p - p^2 = -p^2 - 8p + 48$ Rearranging the terms to make the $p^2$ term first.

$= -1(p^2 + 8p - 48)$ Taking out a common factor of –1

$= -(p + 12)(p - 4)$ Product = –48, sum = 8

Foundation Standard Complex


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## Example 8

Factorise $3x^2 + 8x + 4$.

### Solution

There is no HCF, so we need to split up the middle term $8x$.

Find 2 numbers that have a product of 12 and a sum of 8.

$3x^2 + 8x + 4 = 3x^2 + 8x + 4$ (sum of 8; product of 12 $(3 \times 4)$)

The 2 numbers are $+6$ and $+2$, so we will split $8x$ into $6x$ and $2x$.

$\therefore 3x^2 + 8x + 4 = 3x^2 + 6x + 2x + 4$

$= 3x(x + 2) + 2(x + 2)$ Factorising by grouping in pairs

$= (x + 2)(3x + 2)$ Factorising again

PATH

Video
Factorising quadratic expressions (Advanced)

12.05

## Factorising quadratic trinomials of the form $ax^2 + bx + c$

- Find 2 numbers that have a sum of $b$ and a product of $ac$
- Use these 2 numbers to split the middle term $bx$ into 2 terms
- Factorise by grouping in pairs

## Example 9

Factorise each quadratic expression.

**a** $5k^2 - 12k + 4$ **b** $9m^2 - 9m - 4$ **c** $6t^2 + t - 12$

### Solution

**a** $5k^2 - 12k + 4$

$5 \times 4 = 20$

Find 2 numbers that have a product of 20 and a sum of $-12$.

Since the sum is negative, one of the numbers must be negative.

Since the product is positive, both of the numbers must be negative.

Try $-5, -4$: $-5 \times (-4) = 20, -5 + (-4) = -9$ ✗

Try $-10, -2$: $-10 \times (-2) = 20, -10 + (-2) = -12$ ✓

They are $-10$ and $-2$. Split $-12k$ into $-10k$ and $-2k$.

$5k^2 - 12k + 4 = 5k^2 - 10k - 2k + 4$

$= 5k(k - 2) - 2(k - 2)$

$= (k - 2)(5k - 2)$ Factorising by grouping in pairs

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**b** $9m^2 - 9m - 4$

$9 \times (-4) = -36$

Find 2 numbers with a product of –36 and a sum of –9.

Since the product is negative, one of the numbers must be negative.

Try 12, –3: $12 \times (-3) = -36$, $12 + (-3) = 9$ ✗

Try –12, 3: $-12 \times 3 = -36$, $-12 + 3 = -9$ ✓

They are –12 and 3.

$$9m^2 - 9m - 4 = 9m^2 - 12m + 3m - 4$$
$$= 3m(3m - 4) + 1(3m - 4)$$
$$= (3m - 4)(3m + 1)$$

**c** $6t^2 + t - 12$

$6 \times (-12) = -72$

Find 2 numbers with a product of –72 and a sum of 1.

Try 8, –9: $8 \times (-9) = -72$, $8 + (-9) = -1$ ✗

Try –8, 9: $-8 \times (9) = -72$, $-8 + 9 = 1$ ✓

They are –8 and +9.

$$6t^2 + t - 12 = 6t^2 - 8t + 9t - 12$$
$$= 2t(3t - 4) + 3(3t - 4)$$
$$= (3t - 4)(2t + 3)$$

## Example 10

Factorise each quadratic expression.

**a** $18a^2 - 18a - 8$ **b** $10 - 7x - 12x^2$

### Solution

**a** $18a^2 - 18a - 8 = 2(9a^2 - 9a - 4)$ Taking out the HCF of 2 first

$= 2(9a^2 - 12a + 3a - 4)$ Product = –36, sum = –9

$= 2[3a(3a - 4) + 1(3a - 4)]$

$= 2(3a - 4)(3a + 1)$

**b** $10 - 7x - 12x^2 = -12x^2 - 7x + 10$ Rearranging the terms to make the $x^2$ term first

$= -(12x^2 + 7x - 10)$ Taking out a common factor of –1

$= -(12x^2 + 15x - 8x - 10)$ Product = –120, sum = –7

$= -[3x(4x + 5) - 2(4x + 5)]$

$= -(4x + 5)(3x - 2)$ This can also be written as $(4x + 5)(2 - 3x)$


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**EXERCISE 12.05** ANSWERS ON P. 636

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## Factorising quadratic expressions $ax^2 + bx + c$ U F R

12.05

**1** Factorise each quadratic expression. Look for the highest common factor first. R

EXAMPLE 7

| | | | | | |
|---|---|---|---|---|---|
| **a** | $3m^2 + 9m + 6$ | **b** | $2y^2 + 2y - 4$ | **c** | $5t^2 - 10t - 400$ |
| **d** | $5e^4 + 25e^3 - 120e^2$ | **e** | $x^3 - x^2 - 110x$ | **f** | $4b^2 - 4b - 168$ |
| **g** | $4w^2 + 4w - 48$ | **h** | $3a^3 - 9a^2 - 12a$ | **i** | $2e^2 + 18e + 40$ |
| **j** | $24 - 5t - t^2$ | **k** | $42 + u - u^2$ | **l** | $28 + 3x - x^2$ |
| **m** | $12 - b - b^2$ | **n** | $7k - 12 - k^2$ | **o** | $12x - 35 - x^2$ |

**2** Factorise each quadratic expression. R

EXAMPLE 8

| | | | | | |
|---|---|---|---|---|---|
| **a** | $2x^2 + 11x + 5$ | **b** | $4m^2 + 13m + 3$ | **c** | $5y^2 + 17y + 6$ |
| **d** | $6u^2 + 19u + 10$ | **e** | $2w^2 + 31w + 15$ | **f** | $4e^2 + 15e + 9$ |
| **g** | $8f^2 + 14f + 3$ | **h** | $3d^2 + 5d + 2$ | **i** | $2b^2 + 9b + 7$ |
| **j** | $5y^2 + 16y + 11$ | **k** | $8g^2 + 26g + 15$ | **l** | $6a^2 + 23a + 21$ |

**3** Factorise each quadratic expression. R

EXAMPLE 9

| | | | | | |
|---|---|---|---|---|---|
| **a** | $2y^2 - 11y + 12$ | **b** | $10k^2 - 19k + 6$ | **c** | $6e^2 - 13e + 6$ |
| **d** | $4b^2 - 13b + 3$ | **e** | $6w^2 - 23w + 15$ | **f** | $12f^2 - 25f + 12$ |
| **g** | $15m^2 - 26m + 8$ | **h** | $9x^2 - 12x + 4$ | **i** | $2a^2 - 23a + 45$ |
| **j** | $12y^2 - 35y + 8$ | **k** | $8d^2 - 34d + 21$ | **l** | $4h^2 - 36h + 81$ |

**4** Factorise each quadratic expression. R

| | | | | | |
|---|---|---|---|---|---|
| **a** | $5y^2 - 6y - 11$ | **b** | $4d^2 - d - 5$ | **c** | $2m^2 - 3m - 9$ |
| **d** | $3t^2 - t - 30$ | **e** | $6h^2 - h - 7$ | **f** | $2y^2 - 5y - 12$ |
| **g** | $8a^2 - 2a - 3$ | **h** | $15u^2 - 7u - 4$ | **i** | $9c^2 - 12c - 5$ |
| **j** | $6c^2 - 7c - 24$ | **k** | $20n^2 - 27n - 14$ | **l** | $12x^2 - 7x - 10$ |

**5** Factorise each quadratic expression. R

| | | | | | |
|---|---|---|---|---|---|
| **a** | $5m^2 + 2m - 7$ | **b** | $6g^2 + g - 12$ | **c** | $3p^2 + 4p - 4$ |
| **d** | $7w^2 + 6w - 1$ | **e** | $5y^2 + 14y - 3$ | **f** | $3n^2 + 10n - 8$ |
| **g** | $4b^2 + 9b - 9$ | **h** | $8m^2 + 10m - 3$ | **i** | $3x^2 + 2x - 16$ |
| **j** | $8u^2 + 26u - 15$ | **k** | $24e^2 + 31e - 15$ | **l** | $20h^2 + 11h - 42$ |

**6** Factorise each expression given it is a perfect square. R

| | | | | | |
|---|---|---|---|---|---|
| **a** | $81w^2 - 180w + 100$ | **b** | $4y^2 + 8y + 4$ | **c** | $25h^2 - 40h + 16$ |

**7** Factorise each quadratic expression by first taking out a common factor. R

EXAMPLE 10

| | | | | | |
|---|---|---|---|---|---|
| **a** | $6y^2 + 10y - 4$ | **b** | $6g^2 + 15g - 36$ | **c** | $24e^2 - 28e - 12$ |
| **d** | $8a^2 - 10a - 12$ | **e** | $12u^2 + 20u - 8$ | **f** | $-25q^2 - 5q + 6$ |
| **g** | $-12m^2 + 14m - 4$ | **h** | $20 - h - 12h^2$ | **i** | $24c^2 + 48c + 18$ |
| **j** | $15 + 9w - 6w^2$ | **k** | $12d^2 + 2d - 30$ | **l** | $22x - 12 - 6x^2$ |

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**8** Factorise each quadratic expression. R

| | | | | | |
|---|---|---|---|---|---|
| **a** | $2a^2 + 5a + 3$ | **b** | $12m^2 - 32m + 5$ | **c** | $4x^2 + 11x - 3$ |
| **d** | $7w^2 - 8w + 1$ | **e** | $4h^2 - 7h - 15$ | **f** | $8x^2 - 2x - 3$ |
| **g** | $5r^2 + 26r + 5$ | **h** | $2d^2 - 15d + 7$ | **i** | $6n^2 - 7n - 3$ |
| **j** | $8 - 6m - 9m^2$ | **k** | $3 - 2c - 5c^2$ | **l** | $15g^2 + 19g + 6$ |
| **m** | $15 + 14q - 8q^2$ | **n** | $3x^2 - 13x + 14$ | **o** | $16 - 8d - 3d^2$ |
| **p** | $42 - 10y - 12y^2$ | **q** | $28d^2 - 44d - 24$ | **r** | $100k^2 + 80k + 16$ |

# 12.06 Mixed factorisations

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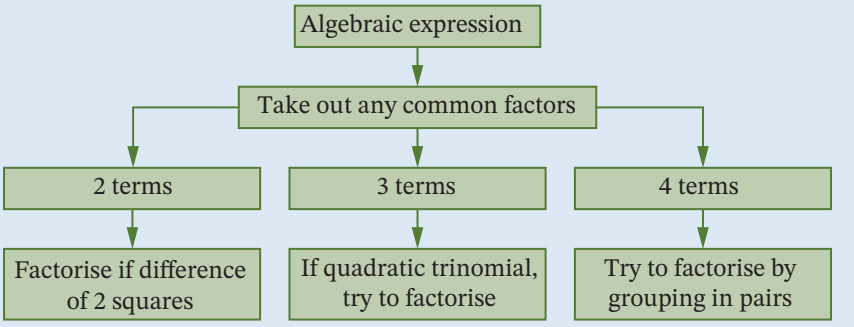
Puzzles
Mixed factorisations

Factorominoes

Products and factors squaresaw

## Factorisation strategies

- Look for any common factors and factorise first.
- If there are 2 terms, try factorising using the difference of 2 squares.
- If there are 3 terms, try factorising as a quadratic trinomial.
- If there are 4 terms, try factorising by grouping in pairs.




## Example 11

Factorise each quadratic expression.

**a** $3a^2 - 27$ **b** $5a^2 + 100$ **c** $20b^2 - 52b + 24$ **d** $d^3 - d^2 - d + 1$

### Solution

**a** $3a^2 - 27 = 3(a^2 - 9)$ Taking out the HCF of 3 first

$= 3(a + 3)(a - 3)$ Difference of 2 squares

**b** $5a^2 + 100 = 5(a^2 + 20)$ 2 terms but not a difference of 2 squares

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**c** $20b^2 - 52b + 24 = 4(5b^2 - 13b + 6)$

$= 4(5b^2 - 10b - 3b + 6)$ Product = 30, sum = –13

$= 4[5b(b - 2) - 3(b - 2)]$

$= 4(b - 2)(5b - 3)$

**d** $d^3 - d^2 - d + 1 = d^2(d - 1) - 1(d - 1)$ Factorising by grouping in pairs

$= (d - 1)(d^2 - 1)$

$= (d - 1)(d + 1)(d - 1)$ Difference of 2 squares

$= (d - 1)^2(d + 1)$

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12.06

## EXERCISE 12.06 ANSWERS ON P. 637

### Mixed factorisations U F R

EXAMPLE 11

**1** Factorise each expression. R

**a** $m^2 - 16m + 64$ **b** $3d^2 - 3d$ **c** $3w^2 - 4w - 15$

**d** $3k - 15 - 5h + hk$ **e** $25y^2 - 64$ **f** $100f^2 - 64$

**g** $q^2 + 3q - 3pq$ **h** $3 + 2g - g^2$ **i** $24n^2 + 44n - 40$

**j** $25r^2 - 1$ **k** $b^3 + b^2 + b + 1$ **l** $4x^2 - 20x + 25$

**m** $4 - d - 5d^2$ **n** $a^3 - a^2 - a + 1$ **o** $8 - 2v^2$

**p** $mn^2 + mnp + 3mn + 3mp$ **q** $2w^2 - 24w + 72$ **r** $36h^2 + 12h + 1$

**2** Factorise each expression. R

**a** $15r^2 - 31rt - 24t^2$ **b** $4d^2 + 4d + 1$

**c** $9g^2 - 36k^2$ **d** $e^3 - 3e^2 - 10e$

**e** $5(p + q)^2 - 125(p - q)^2$ **f** $28x^2 - 7$

**g** $a^2 - b^2 + 4a - 4b$ **h** $c^3 - 2c^2 - 4c + 8$

**i** $6a^2 + 13a - 5$ **j** $y^2 - 3y + 5y - 35$

**k** $18p^2 + 24p + 8$ **l** $1 - 2a - 24a^2$

**m** $9x^2 - 27x + 18x - 54$ **n** $2a^2b - 6ab - 3a + 9$

**o** $2a^2 + 12a + 18$ **p** $25u^2 - 10u + 1$

**q** $4k^2 - 5k - 21$ **r** $48 - 3w^2$

**s** $3 - 27c^2$ **t** $k^3 + 4k^2 - 16k - 64$

**u** $5y^3 - 10y^2 + 15y$ **v** $m^3n - 4mn$

**w** $8 - 2a^2$ **x** $32c^2 - 40c - 12$

**3** Factorise each expression. R

**a** $12e^2 + 13e - 35$ **b** $10y^2 + 19ky - 15k^2$ **c** $16h^4 - 81$

**d** $w^2 - (6 - w)^2$ **e** $d^3 - 16d + 4 - d$ **f** $u^2 - 8u + 16 - k^2$

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# 12.07 Factorising algebraic fractions

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Worksheet
Simplifying algebraic fractions

## Example 12

Simplify each expression.

**a** $\frac{10a+25b}{5}$ **b** $\frac{9y^2-16}{6y+8}$ **c** $\frac{x^2+x}{-4x-4}$ **d** $\frac{t^2-3t+2}{3t^2-5t-2}$

### Solution

**a** $\frac{10a+25b}{5} = \frac{5^1(2a+5b)}{5^1}$

$= 2a+5b$

**b** $\frac{9y^2-16}{6y+8} = \frac{(3y+4)(3y-4)}{2(3y+4)}$

$= \frac{3y-4}{2}$

**c** $\frac{x^2+x}{-4x-4} = \frac{x(x+1)}{-4(x+1)}$

$= -\frac{x}{4}$

**d** $\frac{t^2-3t+2}{3t^2-5t-2} = \frac{(t-2)(t-1)}{(t-2)(3t+1)}$

$= \frac{t-1}{3t+1}$

## Example 13

Simplify each expression.

**a** $\frac{4}{x^2+x}+\frac{3}{x^2-1}$ **b** $\frac{3m-6}{4}\times\frac{8m}{m^2-2m}$ **c** $\frac{d^2+3d+2}{d^2-9}\div\frac{d^2+d}{3d+9}$

### Solution

**a** $\frac{4}{x^2+x}+\frac{3}{x^2-1} = \frac{4}{x(x+1)}+\frac{3}{(x+1)(x-1)}$ Factorising denominators

$= \frac{4(x-1)}{x(x+1)(x-1)}+\frac{3x}{x(x+1)(x-1)}$ Using common denominators

$= \frac{4x-4+3x}{x(x+1)(x-1)}$

$= \frac{7x-4}{x(x+1)(x-1)}$

**b** $\frac{3m-6}{4}\times\frac{8m}{m^2-2m} = \frac{3(m-2)}{4_1}\times\frac{^2\cancel{8}\,m}{m(m-2)}$

$= 6$

**c** $\frac{d^2+3d+2}{d^2-9}\div\frac{d^2+d}{3d+9} = \frac{d^2+3d+2}{d^2-9}\times\frac{3d+9}{d^2+d}$

$= \frac{(d+2)(d+1)}{(d+3)(d-3)}\times\frac{3(d+3)}{d(d+1)}$

$= \frac{3(d+2)}{d(d-3)}$


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## Factorising algebraic fractions U F R

EXAMPLE 12

**1** Simplify each expression. R

**a** $\frac{3x+3y}{3}$ **b** $\frac{5}{10t-10r}$ **c** $\frac{ab-ac}{a^2}$

**d** $\frac{y-1}{1-y}$ **e** $\frac{w^2-16}{w+4}$ **f** $\frac{5d-5w}{d^2-w^2}$

**g** $\frac{(k+5)^2}{k^2-25}$ **h** $\frac{6c^2-6}{2c+2}$ **i** $\frac{am-an+m-n}{m^2-n^2}$

**j** $\frac{y^2+9y+20}{2y+10}$ **k** $\frac{k^2-3k-4}{k^2-16}$ **l** $\frac{16a^2-25c^2}{4a^2-9ac+5c^2}$

**m** $\frac{x^2+4x+4}{x^2-x-6}$ **n** $\frac{1-c-2c^2}{3c^2+2c-1}$ **o** $\frac{ap+4p-2a-8}{2p^2-8}$

**p** $\frac{c^2-9c+20}{cu-4+c-4u}$ **q** $\frac{h^2+3h+2}{2h^2+4h}$ **r** $\frac{8e^2-14e-15}{4e^3-25e}$

EXAMPLE 13

**2** Simplify each expression. R

**a** $\frac{5}{m(m+1)}+\frac{2}{(m+1)(m+2)}$ **b** $\frac{6}{(w+5)(w+3)}-\frac{4}{w(w+3)}$

**c** $\frac{3}{(b+2)(b-1)}+\frac{1}{(b-1)(b-3)}$ **d** $\frac{2}{k^2+k}-\frac{3}{k^2-1}$

**e** $\frac{5}{4h+4}+\frac{3}{h^2+h}$ **f** $\frac{3}{d^2+3d+2}-\frac{4}{d+2}$

**g** $\frac{3}{r^2-36}-\frac{5}{4r+24}$ **h** $\frac{3}{d^2+2d}+\frac{d}{d^2-4}$

**i** $\frac{5}{k^2-3k-4}-\frac{k}{k^2-1}$ **j** $\frac{2}{q^2-1}+\frac{3}{q+1}$

**k** $\frac{4}{a^2-8a+15}+\frac{3}{a^2+4a-21}$ **l** $\frac{2}{y^2+5y+6}-\frac{7}{y^2-y-12}$

**m** $\frac{5}{e^2+3e+2}-\frac{8}{e^2+9e+14}$ **n** $\frac{3}{6h^2-h-2}+\frac{4}{12h^2-23h+10}$

**3** Simplify each expression. R

**a** $\frac{3m+9}{2}\times\frac{4m}{m+3}$ **b** $\frac{5d-10}{3d-9}\times\frac{5d-15}{8d-16}$ **c** $\frac{4}{e+2}\times\frac{e^2+2e}{8e}$

**d** $\frac{3k+6}{5}\times\frac{10k}{k+2}$ **e** $\frac{5h}{3h+9}\times\frac{6h+18}{h^2+h}$ **f** $\frac{4}{a^2-b^2}\times\frac{3a+3b}{8}$

**g** $\frac{r+t}{t^2-r^2}\times\frac{r^2-rt}{5r+5t}$ **h** $\frac{20m+16}{7m-7}\times\frac{7m}{5m+4}$ **i** $\frac{p^2+2p+1}{p^2-1}\times\frac{4p-4}{p^2+p}$

**j** $\frac{y+2}{5y}\div\frac{7y+14}{15y}$ **k** $\frac{5}{x^2-4}\div\frac{15}{2x+4}$ **l** $\frac{4n+8}{n+5}\div\frac{6n+12}{5n+25}$

**m** $\frac{d^2+d}{d+3}\div\frac{6d}{d^2-9}$ **n** $\frac{1}{f^2-6f+9}\div\frac{4}{f^2-9}$ **o** $\frac{3f+6}{f^2+f-6}\div\frac{f^2-2f-8}{f^2-f-12}$





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